“CONFIRMED”
by Rector of Karshi State University
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“____” _____ 2024

FOOD SECURITY AND NUTRITION PROGRAM FOR KARSHI STATE UNIVERSITY STUDENTS FOR 2022-2027

INTRODUCTION

Food safety is a critical factor in maintaining and improving public health, particularly among university students. In the Presidential Decree of the Republic of Uzbekistan, dated November 10, 2020, No. PQ-4887, titled *"On Additional Measures to Ensure the Healthy Diet of the Population,"* it was particularly emphasized that "...failure to follow proper dietary norms and rules, excessive consumption of foods and sweets high in salt, sugar, and fat, as well as insufficient intake of vitamins and minerals, result in delayed physical growth and intellectual development in young people. In adults, this leads to the development of various diseases, such as cardiovascular issues, endocrine disorders, and malignant tumors, which can cause premature death." These conditions can lead to early mortality. In response, a plan of action has been developed to reduce vitamin and mineral deficiencies among children and adolescents and to implement a strategy for rationalizing their nutrition based on the "National Nutrition Research" program. Based on the tasks outlined in the plan, it is of significant scientific and practical importance to study and physiologically analyze the current nutrition of students, identify the relevant issues in this area, develop methodological guidelines for organizing proper nutrition, and create a daily meal plan for them.

As part of this initiative, it is essential to study the nutrition of university students aged 18-29, who are predominantly engaged in mental work. Understanding their nutritional habits and the effects on their mental and physical performance will help guide the development of appropriate dietary plans. This research program, aimed at students of Karshi State University studying full-time, part-time, and in evening classes, seeks to provide scientific, theoretical, and practical assistance in optimizing daily nutrition. The program's goal is to improve student health, promote proper eating habits, and enhance the quality and effectiveness of education.

ASSESSMENT OF THE CURRENT SITUATION OF FOOD SECURITY AND NUTRITION

General Assessment. Food security and proper nutrition are important for students' health, well-being, and academic performance. Karshi State University recognizes the importance of providing students with access to safe, nutritious, and affordable food. Assessing students' dietary habits and the quality of food available on campus helps identify existing problems and develop measures to improve their nutritional well-being. The existing assessment indicates that students' diets may not always meet recommended nutritional standards. Protein and fat intake have been reported to be below recommended levels, while carbohydrate consumption is relatively high. Bread and flour products, fast food, hamburgers, sweets, and carbonated drinks are common parts of students' diets.

At the same time, the consumption of nutritious foods, including meat, dairy products, fruits, vegetables, and eggs, has been reported to be insufficient. These dietary habits highlight the need to promote balanced nutrition and increase students' awareness of healthy food choices.

Access to affordable and nutritious food. Access to sufficient and affordable food is an important part of student well-being. The University recognizes the need to consider students' different financial circumstances and their ability to access nutritious meals. Further assessment of students' needs can help identify those who may require additional support and improve access to affordable food options on campus. The quality and safety of food provided in university cafeterias and other catering facilities are important for protecting students' health. Attention should be given to food storage, preparation, hygiene, and the availability of balanced meals. Providing clear information about food ingredients and nutritional value can also help students make informed choices.

Nutrition awareness and education. Limited awareness of healthy eating may contribute to poor dietary habits. Therefore, nutrition-related workshops, seminars, and educational activities can help students understand the importance of a balanced diet. Cooperation between academic departments, university medical services, and catering facilities can support these efforts. The University also pays attention to the responsible use of food and organic waste. In 2025, food waste generated by campus cafeterias and dormitories amounted to 209.3 tones, which was 108.1 tones less than in the previous year. Of this waste, 18.7 tons were processed and reused as animal feed through granulation. In addition, 18 tons of leaf waste were converted into biomass fertilizer using California red worms and used to maintain the University's green areas. These activities demonstrate efforts to reduce waste and use organic resources more effectively. The assessment identifies several areas that require further attention:

-promoting balanced diets and healthier food choices among students.

-improving access to affordable and nutritious meals.

-increasing awareness of healthy eating and food safety.

-improving the quality and variety of food available on campus.

-continuing to monitor food waste and encouraging its reduction and reuse.

-regularly assessing students' nutritional needs and identifying opportunities for additional support.

Karshi State University recognizes that food security and nutrition are important aspects of student well-being and sustainable campus development. Existing assessments indicate the need to improve dietary habits, promote healthy eating, and ensure access to safe and nutritious food. The University's food waste reduction and organic waste reuse activities also contribute to more responsible resource management. Continued monitoring, awareness-raising activities, and

cooperation between university departments will help strengthen these efforts and support the objectives of Sustainable Development Goal 2 Zero Hunger**.**

KEY ISSUES IMPACTING FOOD QUALITY AND NUTRITION

Body weight imbalance and nutritional health. Previous assessments indicate that a significant proportion of student's experience problems related to body weight. Underweight students account for 21.2%–37.4%**,** while 2.7%–11.3% of students are classified as overweight. In addition, nutrition-related diseases are reported among 9.7%–14.1% of students. These figures highlight the importance of promoting healthy eating habits and improving students' nutritional well-being. Students do not always follow a balanced diet. Bread and flour products, fast food, sweets, and carbonated drinks are common in their diets, while the consumption of meat, dairy products, fruits, vegetables, and eggs may be insufficient. Improving food variety and providing balanced meals in university cafeterias can help address this issue. Some students may lack sufficient knowledge about healthy eating and food safety. They need information about choosing nutritious foods, drinking safe water, using iodized salt, storing food properly, and preparing meals safely. Workshops and educational activities can help students develop healthier eating habits.

Students need to understand the importance of essential nutrients, balanced food portions, and dietary fiber. The University can promote this knowledge through seminars, workshops, and educational materials to help students make better food choices. Improving food quality requires cooperation between university departments, catering facilities, healthcare professionals, and relevant institutions. Regular food quality checks, proper hygiene, and clear information about meals can help improve food safety and student nutrition.

Food waste and sustainable food management.

In 2025, Karshi State University recorded 209.3 tons of food waste from campus cafeterias and dormitories, which was 108.1 tons less than in the previous year. Of this waste, 18.7 tons were processed and reused as animal feed. In addition, 18 tons of leaf waste were converted into biomass fertilizer using California red worms and used in the University's green areas. These activities demonstrate the University's efforts to reduce waste and use resources more responsibly. Topics such as healthy eating, food safety, food quality, and food waste can be included in relevant educational activities, particularly in biology, ecology, and biotechnology. This can help students understand the importance of nutrition and sustainable food practices. The main challenges include unhealthy dietary habits, limited awareness of nutrition, and the need to improve food quality and safety.

At the same time, the University's food waste reduction and organic waste reuse activities demonstrate progress in sustainable resource management.

STRATEGY FOR ENSURING FOOD SECURITY AND ADEQUATE NUTRITION

To address the identified challenges, Karshi State University aims to promote healthy eating, improve food quality, and support the responsible use of food resources through the following measures. Regular assessments should be conducted to understand students' dietary habits and nutritional needs. These assessments can help identify existing problems and develop appropriate measures to improve access to healthy and nutritious food. The University should encourage the provision of safe, fresh, and nutritious food in campus cafeterias. Attention should be given to food variety, hygiene, proper preparation, and the nutritional needs of students. Food safety practices should be improved through proper food storage, preparation, and handling. Catering facilities should follow appropriate hygiene requirements to help ensure that food served to students is safe for consumption.

Food safety and nutrition activities should be implemented and regularly evaluated across the University. Cooperation between university departments and catering facilities can help improve food quality and identify areas for further improvement. The University should continue its efforts to reduce food waste and reuse organic materials where appropriate.

PROPOSED MEASURES TO IMPROVE FOOD QUALITY AND STUDENT NUTRITION

Karshi State University can strengthen research and practical activities related to food quality, nutrition, and food safety through its relevant academic departments. These activities may include studying healthy eating habits, food composition, food quality, and the nutritional needs of students. Cooperation between departments working in biology, ecology, biotechnology, and related fields can support research and help develop practical solutions to food-related challenges. The University should regularly provide students with reliable information about healthy eating, balanced diets, food safety, and responsible food consumption. Information can be shared through the university website, social media, educational materials, seminars, and other communication channels. These activities can help students make informed food choices and better understand the importance of nutrition for their health and academic performance. Practical activities should be organized to raise students' awareness of healthy eating and food safety. Workshops, seminars, awareness campaigns, and educational events can cover topics such as balanced nutrition, food hygiene, safe food storage, and the reduction of food waste. Cooperation between academic departments, student organizations, catering facilities, and relevant institutions can help make these activities more effective.

Staff involved in food preparation, catering, and food management should have appropriate knowledge of food safety, hygiene, storage, and preparation practices. Training sessions can help improve their understanding of safe food handling and the

provision of balanced meals. Academic staff can also contribute by including nutrition and food safety topics in relevant educational and research activities.

University cafeterias should be encouraged to provide a variety of safe, nutritious, and affordable meals. Menus should include different food groups and offer suitable choices to meet students' nutritional needs. Where possible, information about ingredients and nutritional value should be made available to help students choose balanced meals. The University should continue to assess the condition of its catering facilities and identify opportunities to improve food preparation areas, equipment, hygiene, and food storage. Modernization and renovation should be considered where necessary to support food safety requirements and improve the quality of meals served to students. Any improvements should be based on the actual needs of the facilities and available resources. Karshi State University has reported practical activities aimed at reducing food waste and reusing organic materials. In 2025, food waste from campus cafeterias and dormitories amounted to 209.3 tons, a reduction of 108.1 tons compared with the previous year. In addition, 18.7 tons of food waste were processed and reused as animal feed, while 18 tons of leaf waste were converted into biomass fertilizer and used in the University's green areas.

The University should continue monitoring food waste, encouraging responsible consumption, and exploring appropriate methods of organic waste reuse. These measures can support more sustainable food management and reduce the environmental impact of campus activities.

FUNDING SOURCES AND ALLOCATION

Successful implementation of food security and nutrition activities requires appropriate financial planning and the effective use of available resources. Funding priorities should be determined according to the University's needs, institutional budget, and applicable regulations. The University may consider allocating funds within its annual budget to support food safety, nutrition awareness, and

improvements to catering facilities. Depending on available resources, funding may be used for staff training, educational activities, food quality monitoring, equipment upgrades, and the improvement of food preparation and storage conditions. Financial planning should also take into account activities that support the reduction of food waste and the responsible use of organic resources. Access to affordable and nutritious food is an important part of student well-being. The University should assess whether students from low-income families face difficulties accessing regular, balanced meals. Based on the findings and available financial resources, possible support measures may include meal vouchers, targeted subsidies, or other forms of assistance provided in accordance with university policies and applicable regulations.

The University may explore cooperation with public institutions, local organizations, businesses, and other partners to support food security and nutrition initiatives. Where appropriate, grants, project funding, and other lawful sources of financial support may be considered for research, awareness campaigns, food safety improvements, and sustainable food management activities. Financial resources allocated to food security and nutrition activities should be used transparently and in accordance with applicable requirements. Regular monitoring can help assess whether funded activities have achieved their intended objectives and identify areas where further investment may be needed.

EXPECTED OUTCOMES

Implementation of these measures is expected to contribute to:

-increased student awareness of healthy eating and food safety.

-improved access to safe, nutritious, and affordable food.

-better food handling, storage, and hygiene practices in catering facilities.

-stronger cooperation between academic departments and relevant stakeholders.

-continued reduction of food waste and more responsible use of organic resources.

-better identification of students who may need support in accessing nutritious meals.

-progress towards Sustainable Development Goal 2 Zero Hunger through improved food security, nutrition awareness, and sustainable food management.

CONCLUSION

Ensuring food security and promoting healthy nutrition are important for protecting students' health, well-being, and academic performance. Karshi State University recognizes the importance of providing access to safe, nutritious, and affordable food and raising awareness of healthy eating habits. The University's efforts to reduce food waste and reuse organic materials demonstrate its commitment to responsible resource management. Further improvements in food quality, nutrition education, food safety, and access to affordable meals can strengthen these efforts. Cooperation with relevant institutions and the integration of nutrition-related topics into educational activities can also contribute to long-term progress.

Through continued assessment, practical initiatives, and responsible use of resources, Karshi State University aims to strengthen its contribution to Sustainable Development Goal 2 Zero Hunger**.**